

AI Usage Note

Project Name

Postman Collection to Pytest Migrator

AI Tools Used

1. Google AI Studio (Gemini)
2. ChatGPT

What AI Helped With

Architecture Design

AI assisted in:

- Project structure design
- Module separation
- Database planning
- Authentication workflow

Backend Development

AI helped generate:

- Flask routes
- SQLite integration
- Authentication logic
- Session management
- API processing modules

Parser Development

AI assisted in:

- Postman Collection parsing
- Assertion extraction
- Data validation

Conversion Engine

AI assisted in:

- Rule-based conversion logic
- Pytest generation
- JavaScript assertion translation

Frontend Development

AI helped generate:

- Landing page
- Dashboard
- Onboarding wizard
- Authentication pages
- Theme management

Deployment

AI assisted with:

- Render deployment
- Dependency configuration
- Environment setup

What AI Got Wrong

During development AI produced several issues:

Session Management

Problem:

Users were redirected back to login due to iframe cookie restrictions.

Resolution:

Session cookie settings were modified using:

`SESSION_COOKIE_SECURE=True`

`SESSION_COOKIE_SAMESITE="None"`

Frontend Rendering

Problem:

Reports page showed empty recommendation sections.

Resolution:

Database field names did not match frontend template fields.

Mapping logic was corrected.

Navigation Logic

Problem:

Upload and convert buttons redirected users to onboarding.

Resolution:

Navigation routing logic was corrected.

Deployment

Problem:

Render deployment failed because Gunicorn was missing.

Resolution:

Added Gunicorn dependency and redeployed.

Best Prompts Used

Architecture Prompt

Design a complete Flask-based Postman Collection to Pytest Migrator with authentication, dashboard, parser, validator, AI conversion engine, reports, and download center.

Parser Prompt

Create a Postman Collection v2.1 parser that extracts APIs, headers, payloads, variables, and assertions.

Conversion Prompt

Convert Postman JavaScript assertions into executable Pytest assertions using a hybrid rule-based and AI-assisted approach.

Authentication Prompt

Implement secure Flask authentication using SQLite, password hashing, and session-based authorization.

Deployment Prompt

Prepare the Flask application for production deployment on Render using Gunicorn and environment variables.

Human Contributions

The following activities were performed manually:

- Problem analysis
- Feature selection
- Workflow design
- Testing and validation
- Debugging review
- Final deployment
- Documentation review

Conclusion

AI significantly accelerated development by assisting with architecture design, code generation, debugging, testing, deployment guidance, and documentation preparation. Final validation, testing, deployment, and project decisions were performed manually by the development team.